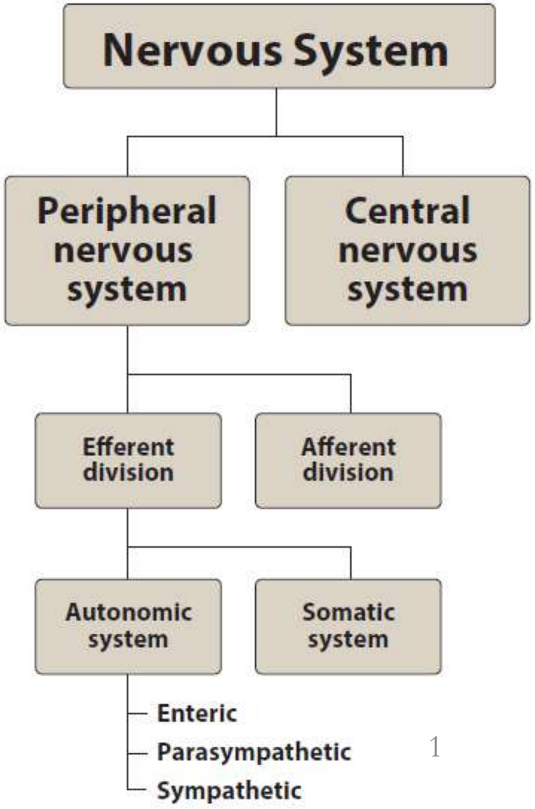
Cholinergic Agonists

PHA 201
Second-year Med

Prof. Huda Alkreathy & Dr. Ahmed Esmat
Department of Clinical Pharmacology,
Faculty of Medicine,
King Abdulaziz University



Objectives

At the end of the lecture, you should be able to:

1. Classify cholinergic agonists and their mechanism of actions.
2. Apply the pharmacological effects of cholinergic agonists for their major clinical uses.
3. Compare the pharmacodynamic differences between direct-acting and indirect-acting cholinomimetic agents.

- **Autonomic physiology is a prerequisite to autonomic pharmacology**

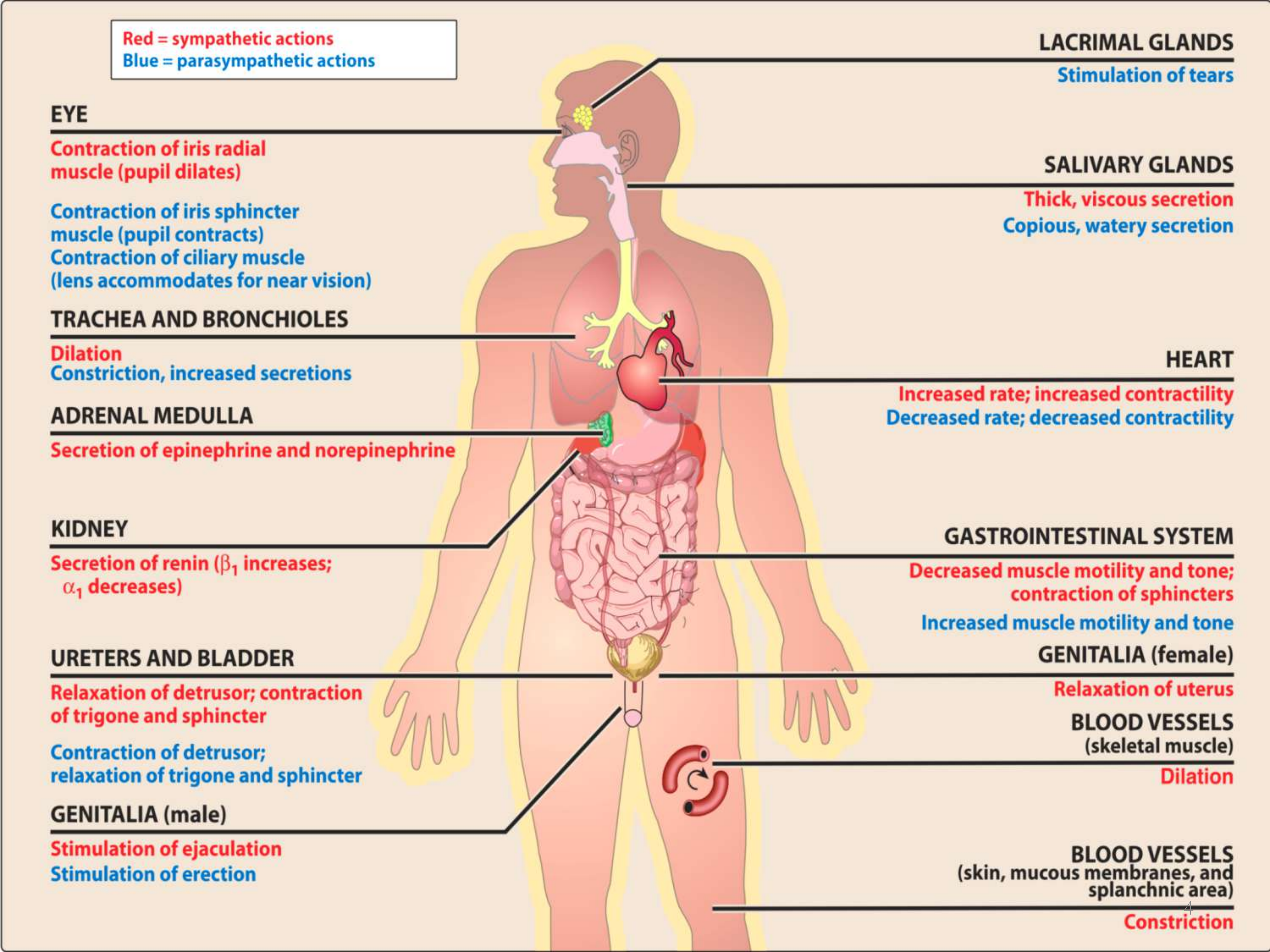
- **You have to read and revise Autonomic physiology**

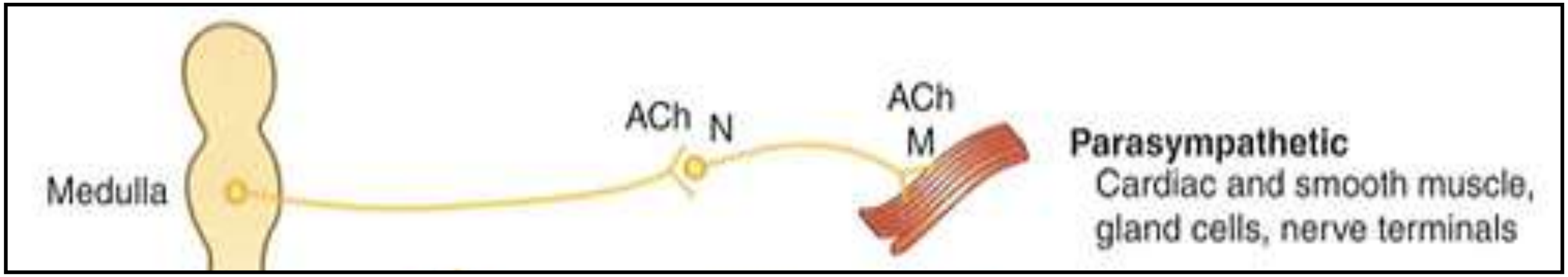
- **Quick review on ANS:**

https://www.youtube.com/watch?v=D96mSg2_h0c

- **Quick review on Cholinergic transmission/receptors/drugs:**

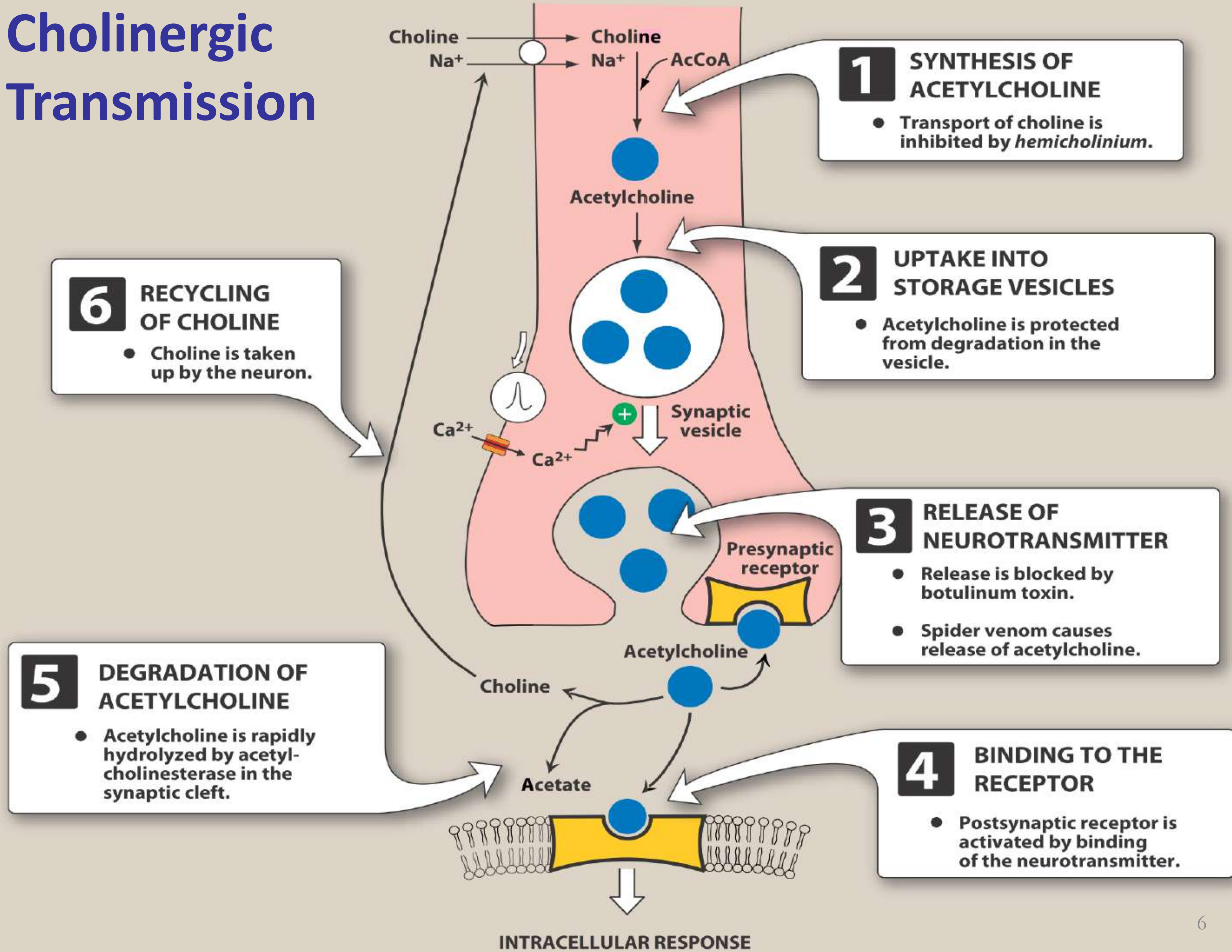
https://youtu.be/EwsVmTOBZrc?si=i9TPzJKTv3KrzkHM


Red = sympathetic actions
Blue = parasympathetic actions
EYE
Contraction of iris radial muscle (pupil dilates)
Contraction of iris sphincter muscle (pupil contracts)
Contraction of ciliary muscle (lens accommodates for near vision)
TRACHEA AND BRONCHIOLES
Dilation
Constriction, increased secretions
ADRENAL MEDULLA
Secretion of epinephrine and norepinephrine
KIDNEY
Secretion of renin (β_1 increases; α_1 decreases)
URETERS AND BLADDER
Relaxation of detrusor; contraction of trigone and sphincter
Contraction of detrusor; relaxation of trigone and sphincter
GENITALIA (male)
Stimulation of ejaculation
Stimulation of erection
LACRIMAL GLANDS
Stimulation of tears
SALIVARY GLANDS
Thick, viscous secretion
Copious, watery secretion
HEART
Increased rate; increased contractility
Decreased rate; decreased contractility
GASTROINTESTINAL SYSTEM
Decreased muscle motility and tone; contraction of sphincters
Increased muscle motility and tone
GENITALIA (female)
Relaxation of uterus
BLOOD VESSELS (skeletal muscle)
Dilation
BLOOD VESSELS (skin, mucous membranes, and splanchnic area)
Constriction



PARASYMPATHETIC NERVOUS SYSTEM

Cholinergic Transmission



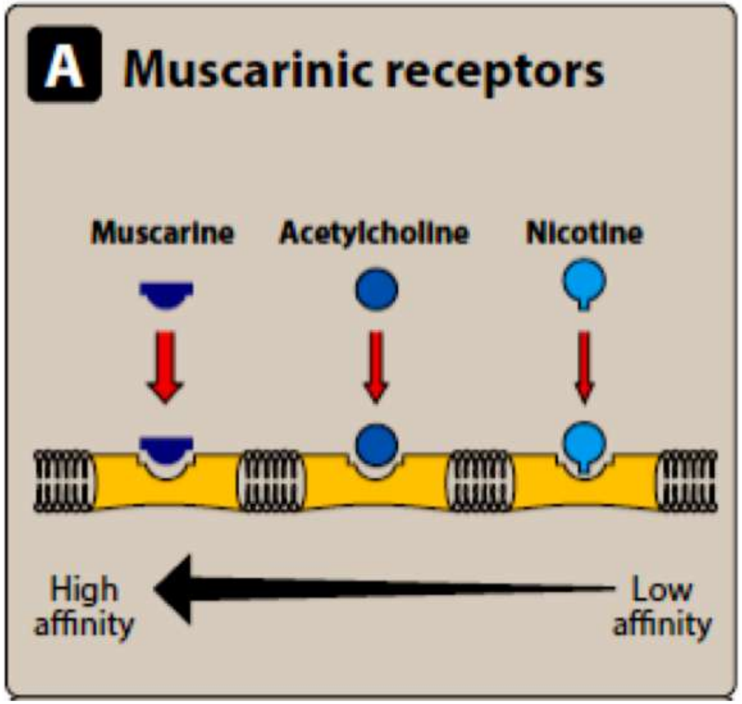
Types of Cholinesterases

True Cholinesterase (AchE)	Pseudo cholinesterase (Pseudo ChE)
Present at terminals of cholinergic fibers and RBCs	Present in plasma
Specific for acetylcholine (Ach)	Non-specific, hydrolyze other choline esters as succinylcholine
Essential for life	NOT essential for life
Slow regeneration within 120 days	Rapid regeneration by the liver

Types of Cholinergic Receptors

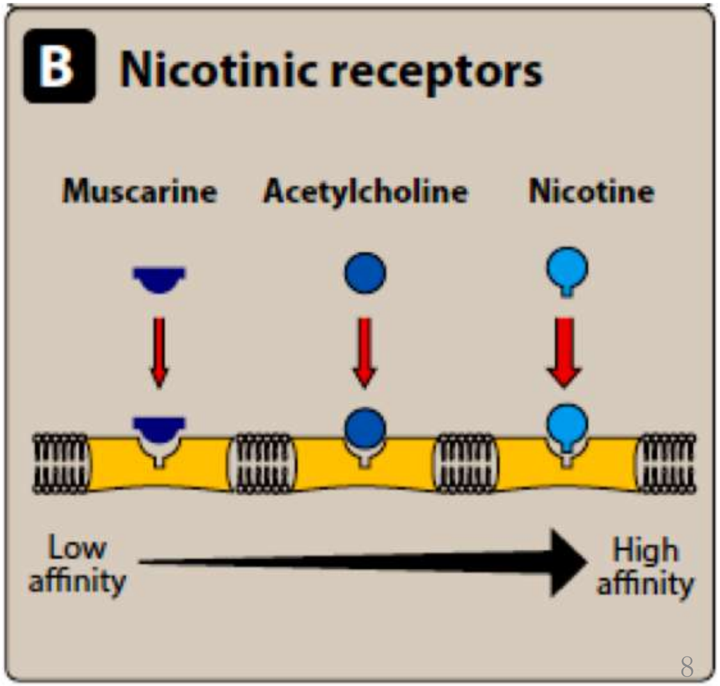
Muscarinic Receptors
(Peripheral Cholinergic R.)

Strong affinity to muscarine
Weak affinity to nicotine



Nicotinic Receptors
(Central Cholinergic R.)

Weak affinity to muscarine
Strong affinity to nicotine

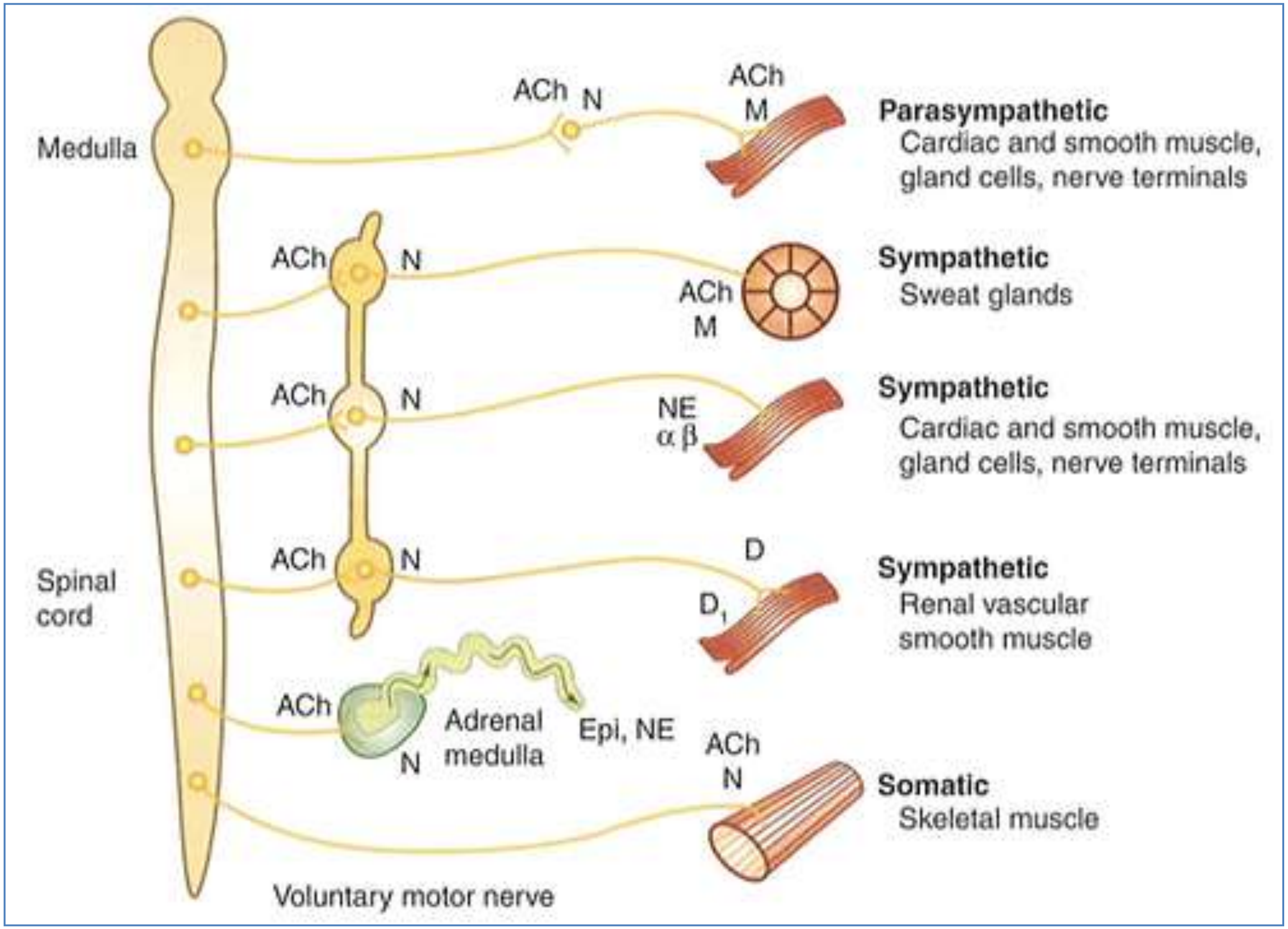


Subtypes of Muscarinic Receptors

M1 receptors (excitatory)	M2 receptors (inhibitory)	M3 receptors (excitatory)
CNS→ excitatory a. Arousal, learning & short-term memory (↓ Ach → Alzheimer). b. In basal ganglia, balance between ACh & DA controls movement. c. In vestibular pathway → stimulate vomiting	• **CNS→ inhibitory** Presynaptic neurons → inhibit ACh release.	**Smooth muscles**→ stimulate wall (bronchi, GIT, urinary) & relaxes sphincters. **Eye** → miosis (stimulate constrictor pupillae) - accommodation (stimulate ciliary muscle).
Stomach→ ↑acid secretion	• **Heart:** a. SAN & AVN → slowing of heart -↓ conduction. b. Atria →↓ contractility -↓ action potential duration .	**Exocrine glands**→ ↑ All secretions (except milk). **Vascular endothelium** → Nitric Oxide release → vasodilation→ ↓ blood pressure.

Nicotinic Receptors (excitatory)

- **(Nn) receptors** in autonomic ganglia, presynaptic on motor nerves & in adrenal medulla → catecholamines release.
- **(Nm) receptors**→skeletal muscle depolarization → contraction (blocked by curare).



Classification of Cholinergic Agonists

CHOLINERGIC AGONISTS

DIRECT ACTING

(M agonists)

- Acetylcholine
- Carbachol (Semisynthetic)
- Bethanechol (Semisynthetic)
- Pilocarpine (Natural)
- Cevemiline (Synthetic)

INDIRECT ACTING

ChE inhibitors

REVERSIBLE

- Edrophonium
- Neostigmine
- Pyridostigmine
- Physostigmine

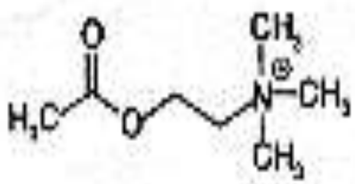
IRREVERSIBLE

- Malathion
- Parathion
- Sarin
- Tabun
- Echothiophate
- Isoflurophate

Acetylcholine (Ach)

The key features of Ach molecule:

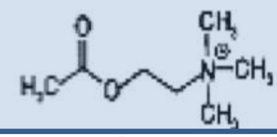
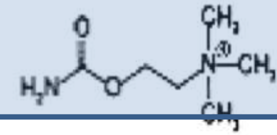
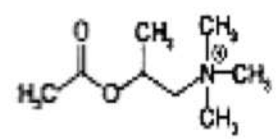
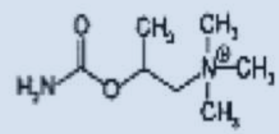
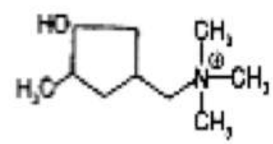
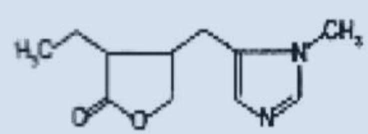
- The quaternary ammonium group
- The ester group

Drug	Structure	Receptor specificity		Hydrolysis by AChE
		Musc	Nic	
Acetylcholine		+++	+++	+++

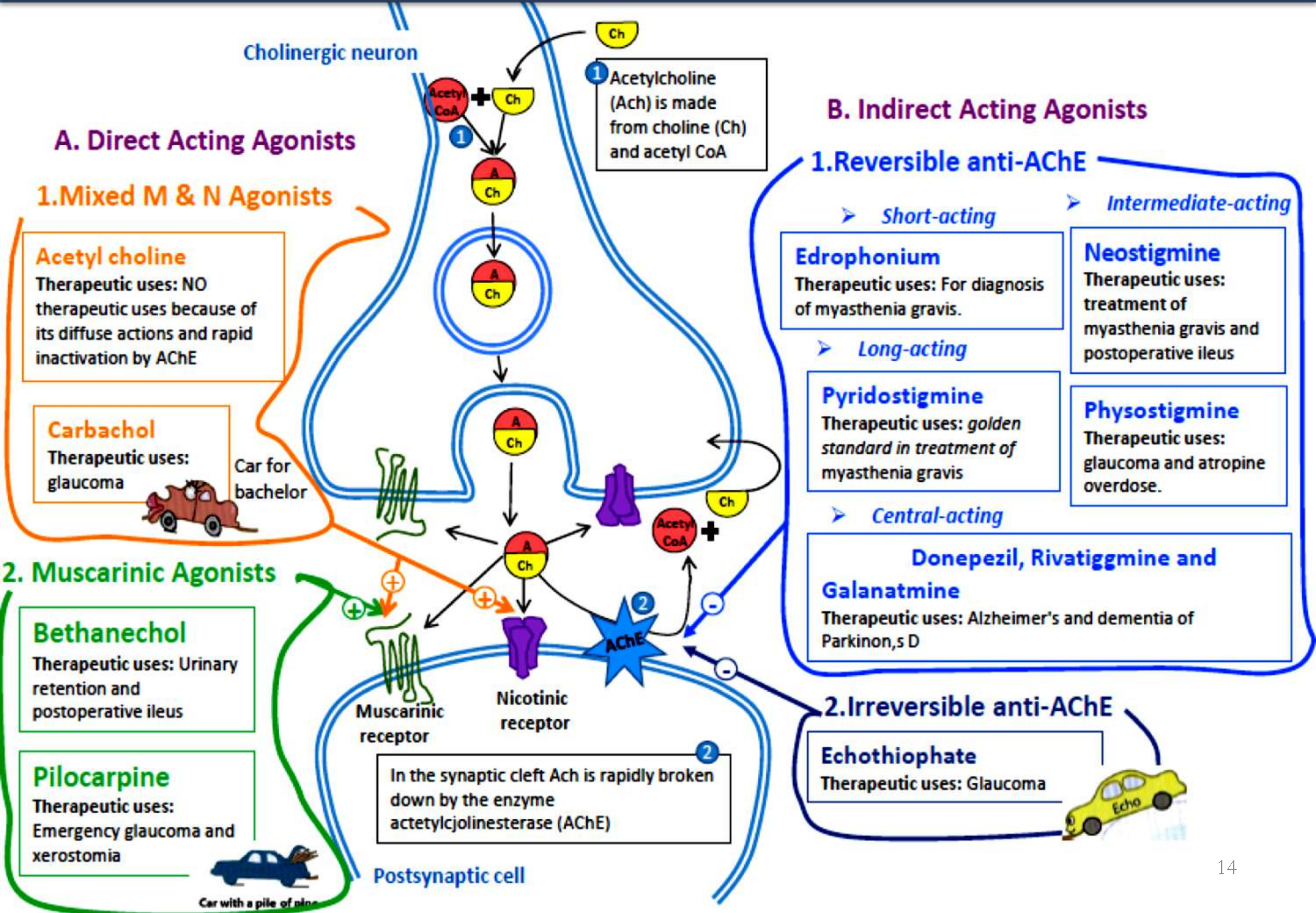
ACh lacks specificity, irregularly absorbed (4ry drug) & rapidly hydrolyzed if given systemically.

Muscarinic Agonists

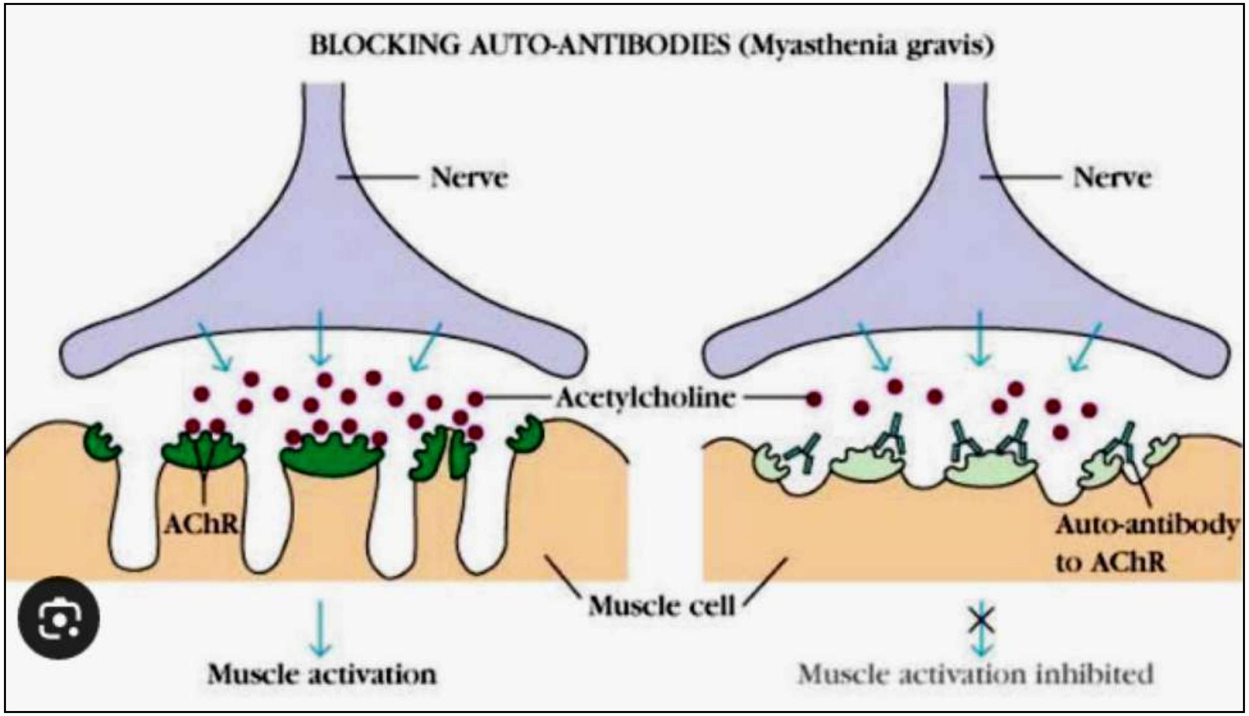
- Substitution of acetyl gp by carbamyl gp → **Carbachol**
- Addition of methyl gp on β carbon → **Methacholine**
- Combining these two modification → **Bethanechol**
- **Muscarine:** Quaternary amine
- **Pilocarpine:** Tertiary amine

Drug	Structure	Receptor specificity: *Musc*	Receptor specificity: *Nic*	Hydrolysis by AChE
Acetylcholine		+++	+++	+++
Carbachol		++	+++	–
Methacholine		+++	+	++
Bethanechol		+++	–	–
Muscarine		+++	–	–
Pilocarpine		++	–	–

Therapeutic uses of Cholinergic Agonists



Myasthenia Gravis overview



- **Autoimmune disorder** affecting **NMJ**; characterized by the presence of IgG auto-antibodies against nicotinic ACh receptors (most common)
- **Pathogenic mechanisms**:
  - Functional blockade of ACh receptors
  - Complement-mediated postsynaptic membrane damage
  - Accelerated receptor internalization and degradation

Myasthenia Gravis Consequences

- **Reduced functional Nm receptors** at the NMJ, leading to decreased end-plate potential → failure to reach threshold
- **Clinical hallmark:** fatigable skeletal muscle weakness
- **Commonly affected muscles:**

  Extraocular (ptosis, diplopia)

  Bulbar and proximal limb muscles
- **Rationale for cholinergic therapy:**

  Increasing ACh availability partially compensates for receptor loss, representing a basis for cholinesterase inhibitors

11.130

11.131

11.130 & 11.131 Myasthenia gravis. The edrophonium (Tensilon) test can be used to confirm the diagnosis. Facial weakness is provoked by repeated facial movements (**11.130**). Edrophonium chloride, a short-acting anticholinesterase, is then given by slow intravenous injection. In myasthenia gravis the facial weakness is rapidly relieved by this test (**11.131**). Objective testing of muscular power elsewhere in the body will reveal similar responses. This test should be always undertaken in hospital where there are resuscitation facilities and with a drawn up syringe of atropine present.

Adverse Effects of Cholinergic Agonists

- Sweating
- Miosis & a spasm of accommodation
- Salivation
- Bradycardia and Hypotension
- Bronchospasm
- Nausea, abdominal pain & diarrhea
- Urinary urgency

A 67-year-old male with a history of benign prostatic hyperplasia (BPH) and post-operative urinary retention presents to the clinic. He complains of an inability to urinate effectively. The physician decides to prescribe a cholinergic agonist to stimulate bladder contraction. Which of the following drugs is most appropriate for this patient?

A) Atropine

B) Bethanechol

C) Neostigmine

D) Pilocarpine

A postoperative patient is experiencing abdominal distension and absence of bowel sounds due to paralytic ileus. The attending physician prescribes a drug that stimulates gastrointestinal motility by inhibiting acetylcholinesterase. Which of the following drugs is most appropriate?

A) Atropine

B) Neostigmine

C) Pilocarpine

D) Pralidoxime

A 75-year-old woman presents with progressive memory loss and cognitive decline. Her physician prescribes a cholinesterase inhibitor to enhance cholinergic transmission in the CNS. Which of the following drugs is most appropriate?

A) Edrophonium

B) Donepezil

C) Atropine

D) Bethanechol

A 50-year-old man takes a high dose of a cholinergic agonist for urinary retention. Shortly after, he develops sweating, salivation, diarrhea, bradycardia, and miosis. Which of the following explains his symptoms?

A) Adrenergic overactivation

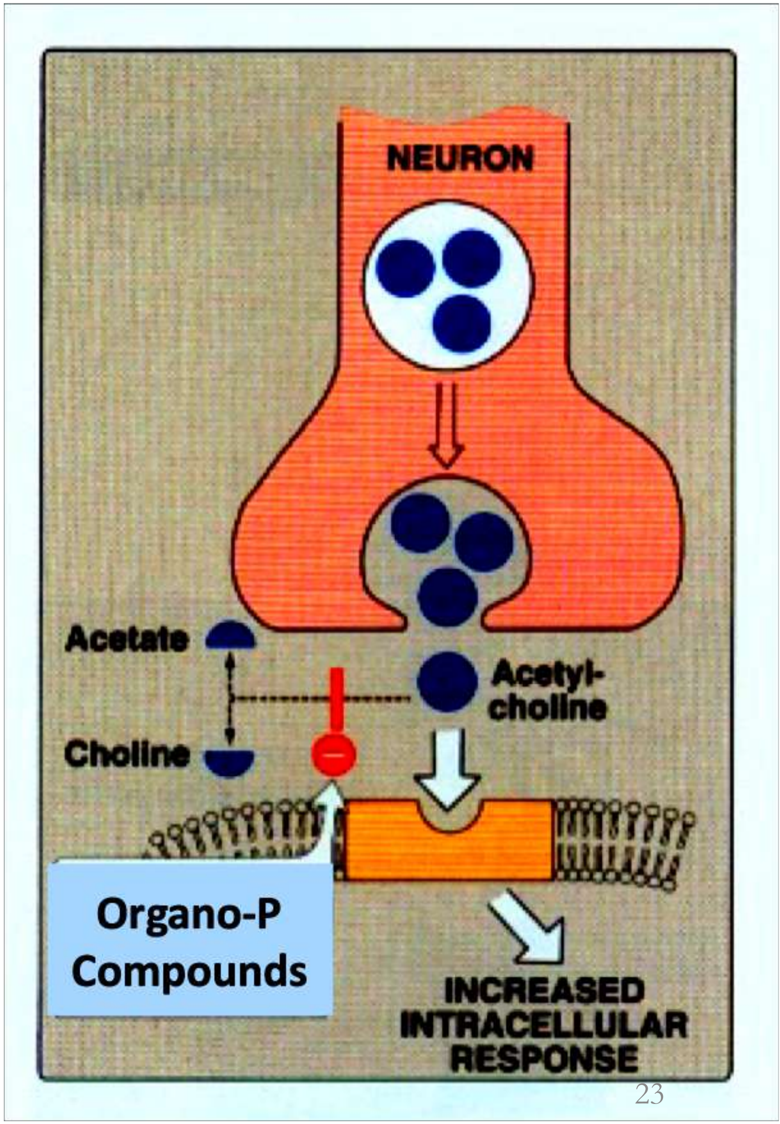
B) Muscarinic receptor activation

C) Nicotinic receptor blockade

D) Sympathetic nervous system activation

Toxicity of irreversible ChE-Inhibitors (Organophosphorus Compounds)

- Organophosphorus compounds are highly lipid-soluble & are well absorbed from all sites & cross BBB .
- Poisoning occurs due to suicide or exposure to drugs during spraying insecticides (Parathion, Malathion) or nerve gases (sarin) during war



Acute Toxicity

- **Excessive muscarinic effects** Defecation, Urination, Miosis, Bronchspasm, Emesis, Lacrimation, Salivation (DUMBELS)
- **Nicotinic effects:** Muscle tremors/fasciculations then flaccid paralysis. Muscle weakness, Adrenal hyperactivity, Tachycardia, Hypertension (MATH)
- **CNS effects**: initial stimulation (may cause convulsions) → followed by depression (coma & respiratory depression).
- **Death is due to respiratory failure**:

- Respiratory center depression.
- Paralysis of respiratory muscles due to persistent depolarization.
- Excessive bronchial secretions with **acute pulmonary edema**.

Chronic Toxicity

- **Malathion→ delayed neuropathy.**

Treatment of Organophosphorus Poisoning

1. **Maintain vital signs** (Respiration & circulation):

   Aspirate bronchial secretions, endotracheal intubation & artificial respiration; IV fluids and BP monitoring

2. **Decontamination** (to prevent further absorption):

   Remove contaminated clothes - wash skin (Na hypochlorite) - gastric lavage.

3. **Symptomatic treatment**

   Atropine (large doses) for CNS & muscarinic effects: 2 -5 mg/ 5 min → full atropinization (until bronchial secretions & wheezes stop).

   Diazepam: for convulsions.

4. **Antidote: Choline esterase reactivators (oximes): PAM** (pralidoxime): Regenerates choline esterase (IV infusion as soon as possible before its ageing).

A 30-year-old farmer is brought to the emergency department spraying his crops. He presents with excessive salivation, sweating, difficulty breathing, and pinpoint pupils. On examination, he also exhibits bradycardia and muscle twitching. Which of the following is the most appropriate initial treatment for this patient?

A. Atropine and pralidoxime

B. Pralidoxime and pilocarpine

C. Intravenoius fluids

D. Activated Charcoal

Important links to view for the next lecture

Glaucoma:

https://www.youtube.com/watch?v=TgjdPgSxeYg

https://www.youtube.com/watch?v=hASPt3hksbA

Power of accommodation:

https://www.youtube.com/watch?v=zLUxn7e8dXI

Common questions

- Which of the following effects is caused by parasympathomimetic agents?

- Which of the following cholinergic receptors are responsible for gastric acid secretion, miosis, learning and memory, bradycardia?

For Your Information

Organ	Sympathetic	R	Parasympathetic	R
Heart	Stimulant	β1	Depressant of atria	M2
Smooth Muscles a- BV	Constriction Dilation (BV Sk. M.) Kidney Vasculature	α β2 D	No innervation (dilation by Nitric Oxide)	M3
b- Bronchi	No innervation (respond to circulating adren.)	β2	Constriction ↑ Secretion	M3 M3
c- GIT -Smooth muscle -Sphincter -Glands	↓ motility Constriction -----	β2 α1	↑ motility Dilation ↑ Secretion ↑Gastric acid Enteric system	M3 M3 M3 M1 M1
d- Urinary Bladder -Smooth muscle -Sphincter	Relaxation Constriction	β2 α1	Contraction Dilation	M3 M3

Organ	Sympathetic	R	Parasympathetic	R
e- Eye:				
* Iris				
-Radial muscle	Contraction	α1		
-Circular muscle			Constriction	M3
*Ciliary Muscle	Relaxation (slight)	β2	Contraction	M3
Glands:				
Salivary Glands	↑ Secretion	α1, β2	↑ Secretion	M3
Lacrimal Glands	No effect	--------	↑ Secretion	M3
Sweat Glands:				
Thermoregulatory	↑ Secretion	M3		
Apocrine (stress)	↑ Secretion	α1		
Metabolic Functions:				
Liver	Gluconeogenesis	β2		
	Glycogenolysis	β2		
Adipose tissue	Lipolysis	β3		
Kidney	Renin release	β1		